

A temperature-free morphological observable that distinguishes non-local from down-gradient heat transport at a melting interface

Abdelrahman Saifelden¹

¹Alexandria Higher Institute of Engineering and Technology (AIET), Alexandria, Egypt

Abstract. Ice and rock scallops migrate downstream as they evolve, a motion classically tied to the phase lag between the wall heat flux and the interface topography: a melting surface migrates downstream when the flux maximum lies downstream of the trough, i.e. when the flux–topography phase lies in $(\pi/2, \pi)$ (Ashton & Kennedy 1972; Gilpin et al. 1980; Hanratty 1981). A local, memoryless, down-gradient (“K-theory”) closure carries no such lag. Freezing a sinusoidal interface $y(x) = \bar{y} + a \sin(Kx)$ and projecting the time-averaged wall flux onto in-phase (amplitude rate β) and quadrature (migration ω_{mig}) harmonics, every coefficient is solver-measured: the conduction amplitude rate is wavelength-independent ($\beta/a \sim K^{+0.13}$), falsifying the K^2 and Mullins–Sekerka $|k|$ ansätze; the in-phase excess is negative at every amplitude and drive (smoothing-limited); and the quadrature grows sub-kinematically ($E_{cos} \propto U^{\approx 0.5}$, friction-velocity controlled), giving a damped, downstream-migrating mode $s = -\beta + i\omega_{mig}$. Swapping only the heat-transport operator at fixed momentum kills or misdirects the quadrature across

5 a **down-gradient closure battery**: machine zero (-1.2×10^{-18}) for the direction-blind uniform eddy diffusivity, and at most 26% of resolved — with the *wrong (upstream) sign* — for flow-aware members (Smagorinsky $K(|S|)$; wall-scaled mixing length $\ell = \kappa d$, cap-insensitive), against -1.1×10^{-4} *downstream* for resolved advection. We then construct a temperature-free observable $I \equiv \tau \cdot c_{mig}/\lambda = |Im(s)|/(2\pi|Re(s)|)$ in which conductivity, latent heat, and basal temperature drop cancel, computable from interface morphology alone; the solver pins $I = O(0.3\text{--}0.9)$ (motion-consistent value ≈ 0.16). A regime-matched

15 **digitization** of Bushuk et al. (2019) — amplitude decay, crest drift, and the adjusting train’s own spacing ($\lambda \approx 28$ mm) read directly off the openly hosted preprint’s panels — confirms the predicted **downstream sign** with net damping, but places the magnitude at $I_{obs} \approx 10$ (estimator battery 5–17: the decay is slow and strongly oscillatory — single-exponential $R^2 = 0.25$, damped-oscillator refit $R^2 = 0.42$), i.e. in the *pre-registered anchoring-mismatch outcome* of our falsification table — as expected for a train whose spacing was formed under the pre-cut flow ($U = 1.00 \rightarrow 0.16$ m s^{−1}) and is far from equilibrium with

20 the drive it is adjusting to. An earlier by-eye bound ($I_{obs} \approx 0.1$, from $\tau \approx 1\text{--}3$ h and a borrowed $\lambda \approx 13$ cm) is superseded: both inputs were wrong by factors of $\gtrsim 5$. We give a field-test protocol and a self-validated reanalysis harness (recovers I to $< 3\%$); the field test is proposed, not performed here, as the raw $h(x, t)$ arrays are not openly available.

1 Introduction

Scalloped erosion/melt patterns are ubiquitous on soluble and meltable surfaces — cave walls, glacier soffits, ice-shelf bases, meteorites, and lab ice (Curl 1966; Blumberg & Curl 1974; Ashton & Kennedy 1972; Bushuk et al. 2019). Their wavelength is fluid-selected through a friction-Reynolds criterion $Re_* = u_* \lambda / \nu \approx 2200$. A robust but under-exploited observation is that

the pattern **migrates downstream** as it evolves — maximum ablation sits downstream of the trough, at flow reattachment (Blumberg & Curl 1974; Gilpin, Hirata & Cheng 1980).

The migration direction and its phase-lag origin are established (Gilpin et al. 1980; Hanratty 1981): downstream migration corresponds to a flux–topography phase in $(\pi/2, \pi)$, and a local, memoryless, down-gradient closure carries no such phase. What is **new here** is not that fact but three things built on it: a phase-separated harmonic decomposition that returns the amplitude and migration rates as independent solver measurements; a direct operator-swap experiment that quantifies how completely a down-gradient closure discards the migration-carrying asymmetry; and a temperature-free morphological observable that makes the phase lag measurable from field geometry alone, without knowing the basal temperature drop or any material constant. The non-local, direction-selecting structure the migration requires is exactly the structure (memory, backscatter) that a local, down-gradient closure cannot represent.

Contributions.

1. A **phase-separated** wall-flux harmonic decomposition that returns the corrected mode $s = -\beta + i\omega_{mig}$ as an *output*, not an ansatz, with the conduction amplitude rate measured wavelength-independent — falsifying the $K^2/|k|$ curvature-smoothing ansätze for the conduction baseline, generalised across amplitude and drive (§3–§4).
2. A **direct operator-swap measurement** quantifying how far down-gradient closures fall short of the known phase-lag requirement: across a closure battery (uniform, Smagorinsky, wall-scaled mixing length) the migration signal collapses to machine zero or flips to the wrong (upstream) sign (§5).
3. A **temperature-free, constant-free morphological observable** $I = \tau \cdot c_{mig}/\lambda$, with an implemented, self-validated reanalysis harness and a regime-matched literature bound, that makes the phase lag measurable from field morphology alone (§6–§8).

2 Method — harmonic decomposition of the wall flux

Freeze $y(x) = \bar{y} + a \sin(Kx)$ ($K = 2\pi n_w/L_x$), time-average the per-column melt flux $m(x) = -\kappa \partial \theta / \partial y$, and project onto the two corrugation harmonics:

$$E_{sin} = 2 \langle e(x) \sin(Kx) \rangle \text{ (in-phase with the **shape** } \rightarrow \text{ amplitude change, } \beta) \quad E_{cos} = 2 \langle e(x) \cos(Kx) \rangle \text{ (in **quadrature** } \rightarrow \text{ pattern **migration**, } \omega_{mig})$$

E_{cos} is the Curl-1966 reattachment/lee signature: zero by symmetry unless the flow carries a direction. Two decompositions are run: the conduction baseline m_{cond} (flow off) for the $\beta \sim K^p$ scaling, and the flow-induced excess $e = m_{flow} - m_{cond}$ versus drive U . Freezing the interface removes boundary motion so the flux is uncontaminated; this is *why* the two coefficients separate cleanly, where a moving-boundary fit conflates them and returns unphysical (both-negative) coefficients.

3 Findings

quantity	measured	ansatz it kills
conduction $\beta/a \sim K^{+0.13}$	wavelength- independent	K^2 curvature and Mullins–Sekerka $ k $ falsified
in-phase flow excess E_{sin}	negative at every driven (a, U)	no autonomous $+\alpha a^{1/2}$ growth — smoothing-limited
quadrature E_{cos}	≈ 0 at $U = 0$, grows $\propto U^{0.5-0.8}$	migration is real and friction-velocity ($u_* \sim U^{1/2}$) controlled, not kinematic ($\propto U$)

Corrected mode: $s(K, U) = -\beta(K, U) + i \cdot \omega_{mig}(U)$, $Re(s) < 0$ always, $Im(s) \propto U^{1/2}$ — a damped, downstream-migrating pattern, not a growth–saturation balance.

60 **Why this counts as a result, not a check.** The corrected mode *fell out of* the decomposition; every coefficient is solver-measured; physical units are pinned independently by the Curl selection law without choosing λ by hand; and the final ratio is constant-free and falsifiable.

4 Generalisation across amplitude and drive

4.1 Amplitude ($a_0/\lambda \in [0.05, 0.40]$). The two structural verdicts are not amplitude artefacts: the driven in-phase excess is
65 **negative at every amplitude** (smoothing throughout, magnitude growing monotonically), and the conduction exponent stays $p \in [-0.27, +0.66]$, far below the K^2 (+2) and $|k|$ (+1) ansätze at every amplitude. *Honest caveat:* strict K -independence ($|p| < 0.6$) and amplitude-flat β/a hold only in the signal-rich regime $a_0/\lambda \gtrsim 0.2$; at shallow amplitude the single-wavenumber conduction signal is at the noise floor (the $K^2/|k|$ falsification is robust regardless). (amplitude_generalization_scan.)

70 **4.2 Drive (U to 6, beyond the swept $[1.5, 3.0]$).** The genuine falsification risk is that strong-drive lee separation adds an *in-phase* component (an emergent growth term the weak-drive sweep would miss). It does not: E_{sin} is **negative at every U** out to $U = 6$, and the migration follows a **sub-kinematic** $E_{cos} \propto U^{+0.48}$ ($\approx 1/2$), far below the kinematic-advection alternative U^1 . *Honest caveat:* migration is non-monotone — it peaks near $U \approx 4.5$ and rolls over by $U = 6$ (the lee structure reaches a limiting form), which *reinforces* the sub-kinematic reading. (drive_window_scan.)

U	in-phase E_{sin}	quadrature E_{cos}
0.0	-5.17×10^{-6} (smoothing)	$+4.9 \times 10^{-8}$ (≈ 0 , parity control)
1.5	-6.23×10^{-5}	-6.59×10^{-5}
3.0	-9.19×10^{-5}	-1.14×10^{-4}
4.5	-8.86×10^{-5}	-1.38×10^{-4}
6.0	-4.81×10^{-5}	-1.19×10^{-4}

5 A direct operator-swap measurement of the closure shortfall

A local, memoryless, down-gradient eddy diffusivity acting on a sinusoid is symmetric under $x \rightarrow -x$: it carries no flow direction and can only return an in-phase (real, damping) response — $E_{cos} \equiv 0$. This is the closure-space statement of the classical phase-lag requirement (Gilpin et al. 1980; Hanratty 1981). The resolved advective flux measured $E_{cos} \neq 0$, so migration requires the flow-direction-selecting, non-local reattachment asymmetry that K-theory discards. The contribution here is to make the shortfall **quantitative**: advancing the **same** momentum field at the **same** drive ($U = 3$, same interface) but transporting heat with the K-theory flux instead of resolved advection measures exactly how far each closure falls below the resolved migration:

heat transport (same $U = 3$, same interface)	E_{cos}	reading
resolved advective $-u \cdot \nabla \theta$	-1.14×10^{-4}	migrates
uniform eddy diffusivity $\nabla \cdot (K \nabla \theta)$	-1.2×10^{-18}	machine zero ($\sim 10^{-14}$ of resolved, $std = 0$)
flow-aware Smagorinsky $K_{eddy}(S)$	$+2.0 \times 10^{-6}$	$\sim 57\times$ below resolved, wrong (upstream) sign
wall-scaled mixing length $\ell = \kappa d$, $K = \ell^2 S $	$+2.96 \times 10^{-5}$	$3.8\times$ below resolved, wrong (upstream) sign ; outer-layer cap $\ell \leq 0.2H$ changes it by $< 0.1\%$

All battery members still **smooth** ($E_{sin} \neq 0$): K-theory can damp but cannot migrate *downstream*. The direction-blind member is exactly parity-protected (machine zero); the flow-aware members inherit only the $|S|$ -magnitude asymmetry of the resolved flow, which pushes their residual quadrature the *wrong way* (upstream, $\leq 26\%$ of resolved) — so no member of the down-gradient family reaches the observed downstream direction at any magnitude (figures/59b_ktheory_closure_battery.json). GPU/CuPy controls (Tesla P100) confirm (a) parity — $E_{cos} \rightarrow +2.1 \times 10^{-6}$ at $U = 0$ vs -8.4×10^{-5} at $U = 3$ ($\sim 40\times$ smaller); and (b) the high- Re_* migration exponent tightens to 0.52 ($nx = 160$, four seeds, scatter $\sim 1-2\%$), on the friction-velocity $\frac{1}{2}$. (scallop_ktheory_control.py.)

This is the morphological analogue of the memory and backscatter signatures that a down-gradient closure omits in *flux* space: the closure's blindness does not merely lose information in the model — it leaves a fingerprint carved into the ice.

6 A constant-free, ΔT -free field test

A Stefan-number-free dimensional bridge ($\rho L v = q$, $q = -k_{th}(\Delta T/L_0)\partial\theta/\partial y$, length anchored by the corrugation wavelength) gives $\tau \sim \lambda^2/\Delta T$ (amplitude e-fold) and $c_{mig} \sim \Delta T/\lambda$ (migration speed). In the product the constants k_{th} , $\rho_i L$ **and** ΔT **cancel exactly**:

$$I \equiv \tau \cdot c_{mig}/\lambda = Im(s)/(2\pi|Re(s)|) \text{ — the dimensionless ratio of migration rate to amplitude rate, basal-}\Delta T\text{-free.}$$

The solver pins $I = O(0.3\text{--}0.9)$ (0.33, 0.68, 0.88 at $n_w = 8, 12, 16$ — wavelength-dependent by a factor ≈ 2.7 , an $O(1)$ morphological number rather than a universal constant).

Decoy warning (a methodological result). $\tau \propto \lambda^2 / \Delta T$ looks like classic curvature/Mullins–Sekerka smoothing, but the λ^2 comes entirely from the conduction length $\leftrightarrow$ time anchoring L_0^2 / κ , **not** from a curvature operator — β/a is K -independent (§3). Fitting $\tau \propto \lambda^2$ in the field would *wrongly confirm* the very curvature ansatz the harmonic probe falsified; the only discriminator is the K -exponent of β/a (≈ 0 here vs $+2$ for curvature), which the dimensional scaling hides. Anyone analysing field scallops should measure the K -exponent, not fit $\tau \propto \lambda^2$.

Field-test protocol. On one scallop train measure λ (crest spacing), c_{mig} (downstream crest displacement between two visits), and τ (amplitude e-fold). Then:

outcome	reading
$I_{obs} \in [0.2, 1.0]$, c_{mig} downstream	consistent with $s = -\beta + i\omega_{mig}$
$c_{mig} \approx 0$ ($Im(s) = 0$)	parity-symmetric closure — K-theory not falsified on the ice (contradicts the solver)
c_{mig} upstream	new physics (e.g. depositional, not melt-limited)
$I_{obs} \gg 1$ or $\ll 0.1$	$\lambda \leftrightarrow u_*$ anchoring mismatch

Because I is ΔT -free, a failure cannot be excused by an unknown basal ΔT ; measuring $c_{mig} + \lambda$ alone pins ΔT , so τ becomes a prediction and the system is over-determined.

7 Literature grounding

- λ (**wavelength**) — **solid**. $Re_* = u_* \lambda / \nu$ selection is mainstream: Blumberg & Curl (1974) ≈ 2200 , Thomas (1979) ≈ 1000 , Hsu–Locher–Kennedy (1979) $\lambda = 3180\nu / u_*$, Thorsness–Hanratty (1979)/Hanratty (1981) band $3100\text{--}6300\nu / u_*$.
- c_{mig} **direction & mechanism** — **solid**. Downstream migration with maximum ablation downstream of the trough at reattachment: Curl (1966), Ashton & Kennedy (1972), Hsu et al. (1979), Gilpin et al. (1980). Gilpin et al.’s linear stability predicts downstream migration when the heat-flux/topography phase shift lies in $(\pi/2, \pi)$ — the same statement as $Im(s) \neq 0$.
- c_{mig} **magnitude and τ** — **sparse**. Cave-scallop surveys report λ only; migration speed and amplitude decay time are essentially never tabulated together for one train.

Two derived consistency relations with the turbulence closure (Paper 1). The migration quadrature $Im(s)$ that distinguishes a real closure is the imaginary part of a single complex transport admittance $Z = a_r + ia_i$, whose real part is the amplitude-rate / backscatter of the closure (Paper 1; Kraichnan 1976) and whose imaginary part is this scallop migration; a down-gradient eddy diffusivity is the projection $Z \mapsto \max(Re Z, 0)$, which zeroes migration and backscatter *together*. Because the flux response is moreover causal, $Re Z(k)$ and $Im Z(k)$ are Kramers–Kronig (Hilbert) conjugates: any *scale-dependent* eddy viscosity — which every real closure (spectral eddy viscosity, Smagorinsky $\propto |S|$) is — **forces** a nonzero migration, so

K-theory’s pairing of a scale-dependent $\text{Re } Z$ with $\text{Im } Z \equiv 0$ is acausal (an $O(1)$ Kramers–Kronig residual). The measured migration $I = |\text{Im } Z|/(2\pi|\text{Re } Z|)$ is therefore not an incidental fact but the causality-mandated shadow of the same operator that carries backscatter in the turbulence closure — a derived link, verified on the admittance model (discrete Kramers–Kronig reconstruction of $\text{Im } Z$ from $\text{Re } Z$ to 5.3%).

130 8 The one open gate — and a regime-matched bound

Sign caveat. Real ice ripples *grow* ($\text{Re}(s) > 0$, the Ashton–Kennedy/Gilpin/Hanratty moving-boundary instability), whereas the frozen-interface excess here is decay-only ($\text{Re}(s) < 0$) by construction. The migration $\text{Im}(s)$ is the robustly shared, K-theory-falsifying piece; matching the *growing* branch needs an interface-growth mechanism (a Röthlisberger opening) this smoothing-only solver does not contain.

135 **The migration transfers to a moving boundary.** We close the part of this gate the solver *can* reach. Seeding a single mode on the interface and advancing it under the Stefan melt feedback ($\rho Lv = q$), the co-evolving interface tracks a **coherent damped-migrating eigenmode**: $\log|Z|$ and $\arg(Z)$ of the fundamental shape mode are both linear in time ($R^2 \geq 0.91$ for amplitude and phase, two seeds $\times$ two Stefan numbers). It **decays** ($\text{Re}(s) < 0$ at every St — the decay-only branch is *physical*, not an artefact of freezing) and **migrates downstream**, with $\text{Im}(s)$ tracking the frozen-probe quadrature $E_{\cos}$ (predicted
140 $\text{Im} = E_{\cos}/(aSt) \approx -0.43$ vs measured -0.40 at $St = 10^{-3}$, $\sim 5\text{--}10\%$). The observable $I = |\text{Im}|/(2\pi|\text{Re}|) \approx 0.15\text{--}0.17$ is **St -invariant** (the rate scales as $1/St$, so $\text{Re} \cdot St$ and $\text{Im} \cdot St$ are constant to $\sim 3\%$), confirming I is a well-defined, normalization-free property of the mode. The migration — the closure-distinguishing quadrature — therefore **survives interface motion**. The total moving-interface decay runs $\sim 1.75\times$ faster than the flow-off conduction estimate (turbulence adds effective smoothing), so this motion-consistent $I \approx 0.16$ sits *below* the frozen flow-off band; the digitized Bushuk adjustment-train reading (below)
145 lies *above* both — an anchoring mismatch of the observed train, not a solver discriminant. The *growing* branch ($\text{Re}(s) > 0$) is still out of reach here. (`scallop_moving_boundary_check.py`, `subglacial/moving_boundary_check.json`.)

A regime-matched real-data bound. Bushuk et al. (2019, *JFM* 873) track an evolving ice–water interface $h(x, t)$ at sub-mm/15 Hz through transition→equilibrium→**adjustment**. Their adjustment regime (after the drive was cut $U = 1.00 \rightarrow$
150 0.16ms^{-1}) **damps and migrates downstream simultaneously** — the exact analogue of the frozen probe ($s = -\beta + i\omega_{\text{mig}}$, $\text{Re}(s) < 0$, $\text{Im}(s) \neq 0$), so the sign caveat does not apply. We **digitized the adjustment-regime panels directly** from the openly hosted preprint (sha256-pinned; `glaciers/bushuk_fig3_digitize.py`: content-located page, tick-label-anchored piecewise calibration, density-tracked amplitude, slanted-comb crest fit; artifacts `figures/58b_bushuk_digitized.json/.png`). The digitized kinematics are: amplitude $11.0 \rightarrow 9.2$ mm over ~ 300 min (early/late-window quartiles) — slow
155 and **strongly oscillatory**, so τ comes from a fit-model battery: the physically-motivated damped-oscillator refit $A_0 e^{-t/\tau} [1 + b \cos(2\pi t/T + \phi)]$ gives $\tau = 2.6 \times 10^3$ min (68% CI $2.1\text{--}3.4 \times 10^3$; $T \approx 69$ min, $b \approx 0.03$, $R^2 = 0.42$), bracketed by envelope quartiles (1.2×10^3 , conservative) and the log-linear fit (2.9×10^3 , $R^2 = 0.25$; a two-exponential control degenerates, $R^2 = 0.30$); crest drift $c = 0.11$ (their published mean, anomalous periods excluded) to 0.17 mm min^{-1} (all-period comb);

and the adjusting train’s own spacing $\lambda = 27.9$ mm (6 crests, spacings 21–31.5 mm). An earlier by-eye reading ($\tau \approx 1\text{--}3$ h, borrowed equilibrium-band $\lambda \approx 13$ cm, giving $I_{obs} \approx 0.10$) is **superseded** — its τ was $\geq 5\times$ too fast and its λ was not this train’s spacing; the two errors happened to cancel toward the band:

$I_{obs} = \tau \cdot c_{mig} / \lambda \approx 10$ (preferred damped-oscillator τ with the published c ; estimator battery 4.9–17): **downstream sign and net damping as predicted**, but a magnitude 6–20 $\times$ above the frozen flow-off band $[0.33, 0.88]$ and 30–110 $\times$ above the motion-consistent solver value 0.16 — i.e. squarely in the **pre-registered** $I_{obs} \gg 1$ anchoring-mismatch row of the §7 outcome table. That is the physically expected bin for this regime: the adjusting train retains the spacing formed under the *pre-cut* drive ($U = 1.00$ m s $^{-1}$) while damping under a 6 $\times$ weaker flow, so its λ is anchored to the wrong u_* — exactly the mismatch the table names — and its oscillatory amplitude makes a single-eigenmode τ estimator-dependent (the dominant spread in I_{obs}). The **sign structure** — the closure-falsifying content — is confirmed; the **magnitude** question now rests on the raw-array pin: $h(x, t)$ through the *equilibrium* regime (where the solver band applies), read by the §6 harness, would replace both τ and c with the complex rate of the same mode. Candidate datasets beyond Bushuk’s arrays (on request; co-authors D. Holland / A. Stern): Weady et al. (2022, PRL 128, 044502) interface extractions, and Cohen/Berhanu dissolution-scallop time-lapses (growth regime; sign caveat applies).

The pin. The one missing input is Bushuk’s underlying $h(x, t)$ arrays, available from the authors on request (mitchell.bushuk@noaa.gov); the published JFM supplement is a parameter table only, not the interface time series, and no open data deposit exists. (The *equilibrium* regime cannot substitute for the raw arrays: there $Re(s) \approx 0$ by definition — the amplitude is statistically steady — so I is not readable from a published equilibrium panel; the adjustment regime is the only one whose figure exposes both rates, which is also why the raw $h(x, t)$ remains the true pin.) The §6 ratio is computed from morphology alone, and the reanalysis harness is **already implemented and self-validated**: `harmonic_mode_rate(x, t, H)` rFFTs each frame, tracks the dominant mode $a_k(t)e^{i\phi_k(t)}$, fits $Re(s) = d/dt \ln a_k$, $Im(s) = d\phi_k/dt$, and returns I plus the downstream sign; on a synthetic damped, downstream-migrating train with **known** ($Re(s), c_{mig}, \lambda$) it recovers all three to $< 2\%$ and I to $< 3\%$. It ingests Bushuk’s arrays directly — no further code, just the data. (`scallop_field_test.py`.)

9 Discussion and scope

This is a solver-verified, quantitative characterization of the closure: the corrected mode, the operator-swap shortfall, and the temperature-free ratio are all self-consistent and GPU-reproducible. **Dimensionality.** All probes here are 2-D ($x\text{--}y$), while real scallops are three-dimensional cups with spanwise structure. The *structural* verdicts are dimension-robust by symmetry: a local down-gradient closure carries no flow direction in any dimension, so its quadrature vanishes (or, flow-aware, misdirects) in 3-D exactly as in §5 — the parity argument never references dimensionality. The *numbers* (the $U^{\approx 0.5}$ quadrature growth, the $K^{+0.13}$ conduction exponent, the solver band for I) are 2-D measurements; their 3-D counterparts are part of the field-pin step, not assumed. It is **not yet validated against field scallops** — §8 is precisely why that test is cheap (no ΔT needed).

The banked claims are the qualitative three — $Im(s) \rightarrow 0$ at $U = 0$, strictly sub-linear migration, exponent $\rightarrow \frac{1}{2}$ at high Re_* (central value 0.52) — robust to the $\sim \pm 0.1$ scatter from spin-up/seed/ U -grid. The solver-side moving-boundary measurement (§8) is now done — the migration transfers to an actually-moving interface and I is motion-invariant — so what a field dataset would still pin is the migration exponent’s precise value and the field I_{obs} from real $h(x, t)$.

195 10 Reproduce

```
python glaciers/scallop_amplitude_harmonics.py # $3-$4; writes figures/56_*.json
python glaciers/scallop_ktheory_control.py # $5 keystone control (CPU; xp=cupy for GPU)
python glaciers/scallop_field_test.py # $6/$8; figures/58_*.json (harness self-check + super
python glaciers/bushuk_fig3_digitize.py # $8; figures/58b_*.json (digitized exp-1b kinematic
200 python glaciers/scallop_moving_boundary_check.py # $8 moving-boundary migration transfer + I
pytest glaciers/tests/test_scallop_amplitude_harmonics.py \
    glaciers/tests/test_scallop_ktheory_control.py \
    glaciers/tests/test_scallop_field_test.py
```

Code and data availability. All scallop-migration and harmonic-decomposition results regenerate from the scripts in the reproduce section;
205 the field-test harness and its synthetic validation data are in the public repository at <https://github.com/abdh0saifelden2-debug/K>.

Author contributions. A. Saifelden conceived the study, performed all computations and analysis, and wrote the manuscript.

Competing interests. The author declares that there are no competing interests.

References

- Ashton, G. D., & Kennedy, J. F. (1972). Ripples on underside of river ice covers. *Journal of the Hydraulics Division (ASCE)* 98, 1603–1624.
210 doi:10.1061/JYCEAJ.0003407
- Blumberg, P. N., & Curl, R. L. (1974). Experimental and theoretical studies of dissolution roughness. *Journal of Fluid Mechanics* 65,
735–751. doi:10.1017/S0022112074001625
- Bushuk, M., Holland, D. M., Stanton, T. P., Stern, A., & Gray, C. (2019). Ice scallops: a laboratory investigation of the ice–water interface.
Journal of Fluid Mechanics 873, 942–976. doi:10.1017/jfm.2019.398
- 215 Curl, R. L. (1966). Scallops and flutes. *Transactions of the Cave Research Group of Great Britain* 7, 121–160.
- Gilpin, R. R., Hirata, T., & Cheng, K. C. (1980). Wave formation and heat transfer at an ice-water interface in the presence of a turbulent
flow. *Journal of Fluid Mechanics* 99, 619–640. doi:10.1017/S0022112080000791
- Hanratty, T. J. (1981). Stability of surfaces that are dissolving or being formed by convective diffusion. *Annual Review of Fluid Mechanics*
13, 231–252. doi:10.1146/annurev.fl.13.010181.001311
- 220 Hsu, K.-H., Locher, F. A., & Kennedy, J. F. (1979). Forced-convection heat transfer from irregular melting wavy boundaries. *Journal of Heat*
Transfer 101, 598–602. doi:10.1115/1.3451043
- Mullins, W. W., & Sekerka, R. F. (1964). Stability of a planar interface during solidification of a dilute binary alloy. *Journal of Applied*
Physics 35, 444–451. doi:10.1063/1.1713333
- Thomas, R. M. (1979). Size of scallops and ripples formed by flowing water. *Nature* 277, 281–283. doi:10.1038/277281a0
- 225 Thorsness, C. B., & Hanratty, T. J. (1979). Mass transfer between a flowing fluid and a solid wavy surface. *AIChE Journal* 25, 686–697.
doi:10.1002/aic.690250415